BLE Specs

Birdie PRO

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Purpose of Document

This document provides detailed information about the GATT services and characteristics available on the Birdie Pro BLE device, including UUIDs, value sizes, characteristic properties, and interpretation notes.

1. Generic Access Service (UUID: 1800)

1.1 Device Name

UUID: 2A00
Value Size: Variable
Properties: Read

1.2 Appearance

UUID: 2A01
Value Size: 2 bytes
Properties: Read

2. Birdie Environmental Service (UUID: 4c9b9a7b-370d-473b-8109-f4dfeb661012)

2.1 Carbon Dioxide Concentration

UUID: 8fd61dc9-bb74-48c8-924a-d95c57c2f42f
Value Size: 2 bytes
Properties: Read, Notify
Interpretation: uint16_t, ppm

2.2 Temperature

UUID: 0b91651c-cadb-490c-92ca-45cc8f40f1c9
Value Size: 2 bytes
Properties: Read, Notify
Interpretation: int16_t, millidegrees Celsius (e.g., 22500 = 22.5°C)

2.3 Humidity

UUID: 0842ee99-4d87-40c5-89c2-8e5b23f37c59
Value Size: 2 bytes
Properties: Read, Notify
Interpretation: uint16_t, milli-%RH (e.g., 45678 = 45.678% RH)

2.4 Birdie state

UUID: 6fdbae35-d241-4a74-bbc3-992010010601
Value Size: 1 byte
Properties: Read, Notify
Interpretation: uint8_t ,**0-**BIRDIE_UP, **1-**BIRDIE_DOWN **2-**BIRDIE_COOLDOWN

3. Birdie Battery Service (UUID: 30b69cc2-03ce-4e58-aa95-b8bffffb4694)

3.1 Battery Level

UUID: 4769d60c-49b1-4e4b-bc0c-149c5e757bdd
Value Size: 1 byte
Properties: Read, Notify
Interpretation: 0 = Full, 1 = 75%, 2 = 50%, 3 = 25%, 4 = Empty

4. Configuration Service (UUID: 1db703c7-2354-4ffb-bab5-2ae18e414791)

4.1 Recalibration Threshold

UUID: f7ac6b7e-bb41-4bc3-9fc8-35f73b90f157
Value Size: 2 bytes
Properties: Read, Write
Interpretation: uint16_t, ppm

4.2 Bad IAQ Period

UUID: b2035c1a-07a9-42ce-a2b7-d97c6a34784a
Value Size: 1 byte
Properties: Read, Write
Interpretation: uint8_t, minutes

4.3 IAQ Threshold

UUID: 5f46fade-19a9-4dd5-991e-22729e3ebc44
Value Size: 2 bytes
Properties: Read, Write
Interpretation: uint16_t, ppm

4.4 Cool Down Period

UUID: 2f0c3872-c2be-4416-9610-14a103ed1fc7

Value Size: 1 byte
Properties: Read, Write
Interpretation: uint8_t, minutes

4.5 Good IAQ Period

UUID: 07efb304-17a3-4b77-9adf-777195a3b3ec
Value Size: 1 byte
Properties: Read, Write
Interpretation: uint8_t, minutes

4.6 Firmware Revision String

UUID: 8798982b-8e1a-4102-ab20-b8f1b6a75495
Value Size: 3 bytes
Properties: Read
Interpretation: Byte[0] = major, Byte[1] = minor, Byte[2] = patch

4.7 Hardware Version

UUID: a418b935-1e12-4e2a-9e30-81c91ea6b5a0
Value Size: 3 bytes
Properties: Read
Interpretation: Byte[0] = major, Byte[1] = minor, Byte[2] = patch

4.8 Averaging window size

UUID: 0fdde7ea-1fe8-48e0-b6c9-9516b3af0233
Value Size: 1 byte
Properties: Read, Write
Interpretation: uint8_t, last N samples used for averaging, MAX value is 12

5. Current Time Service (UUID: 6a2a22df-f59d-4cea-aa62-ac2ee0507ed2)

This is a write-only characteristic used for device time synchronization.

5.1 Current Time

UUID: 1dfdc913-9801-4529-bef4-c5ceb0affeb9
Value Size: 4 bytes
Properties: Write Interpretation: uint32_t, Unix Timestamp (seconds since epoch in UTC Little Endian notation)

6. Birdie Data Log Service (UUID: 9daf8c05-ec2b-448d-8c83-ba19f447e9c0)

This service facilitates the transfer and retrieval of historical sensor data.

6.1 Log Data Channel

UUID: 393611dd-4e5f-4023-95ae-623dce1d7242
Value Size: 14 bytes per entry
Properties: Read, Notify
Interpretation: Used for both single index read result and log streaming notifications.

6.2 Log Index Write

UUID: c451887b-6516-48d8-9cb5-4c360bee29b0
Value Size: 2 bytes Properties:
Write Interpretation: `uint16_t` (**0-8157**) - index of the log entry to read.

6.3 Log Entry Data Structure (14 bytes):

Offset	Size	Type	Field	Description
0	2 bytes	`uint16_t`	**CO2**	CO2 concentration in ppm.
2	4 bytes	`int32_t`	**Raw Temp**	Temperature in millidegrees Celsius.
6	4 bytes	`int32_t`	**Raw Hum**	Humidity in milli-%RH.
10	4 bytes	`uint32_t`	**Timestamp**	Unix Timestamp (seconds). 0 if log is empty.

7. Silicon Labs OTA Service (UUID: 1D14D6EE-FD63-4FA1-BFA4-8F47B42119F0)

7.1 Silicon Labs OTA Control

UUID: F7BF3564-FB6D-4E53-88A4-5E37E0326063
Value Size: Variable
Properties: Write

8. Ota mode scan response

When the device enters OTA (bootloader) mode, it advertises as "Birdie Pro-XX-XX-OTA"
where XX-XX are the last two octets of the BLE MAC address in uppercase hex.
And has a scan response,the scan response contains two AD structures (23 bytes total):

Field	Offset in scan response	Length	AD Type	Description
TX Power	0-2	3 bytes	`0x0A`	TX power level in dBm (integer)
HW Version	3-22	20bytes	`0x21` (Service Data 128-bit UUID)	UUID `a418b935-1e12-4e2a-9e30-81c91ea6b5a0` followed by 3 bytes: `[major, minor, patch]`

Byte Offset of full scan response	Value	Field
0	0x02	AD length (2 bytes follow)
1	0x0A	AD type: TX Power Level
2	TX_POWER / 10	TX power value (dBm)
3	0x14 (20)	AD length (20 bytes follow)
4	0x21	AD type: Service Data - 128-bit UUID
5-20	16 bytes	UUID a418b935-1e12-4e2a-9e30-81c91ea6b5a0 in little-endian
20	hw_major	HW version major
21	hw_minor	HW version minor
22	hw_patch	HW version patch

The scan response contains two AD structures (23 bytes total):

1. Look for AD type 0x21 (Service Data - 128-bit UUID)
2. Check if the UUID matches the hardware version characteristics UUID (a418b935-1e12-4e2a-9e30-81c91ea6b5a0)
3. Read the 3 bytes after the UUID as hw_major, hw_minor, hw_patch at the defined offsets